

BUSINESS INTELLIGENCE

Who reaches Chile's public hospitals with private insurance?

Severity and admission profile of Isapre/Particular vs FONASA discharges, GRD 2020 - 2024.

From W1 to here...

- ✗ **Candidate A — Emergency care by age group and facility type**
- ✓ **Candidate B — Public vs. private gap in hospital case complexity**

Changes done after initial W1's feedback:

- Selected Candidate B and loaded the complete GRD files for 5 years (2020-2024, ~700K-1M rows/year), not a single record.
- Computed 3 KPIs with real values across all 5 years (severity share, average case weight, urgent-admission share).
- README completed; notebook restructured, executed top-to-bottom with saved outputs.
- Data-quality checks (Five Cs) run directly against the raw files: encoding inconsistency across years, malformed-line count, population-consistency check for the 2022 "EXTERNO" file.

CI's actual question:

“ Among privately insured patients who do end up in the public network, is their severity profile lower, similar or higher than FONASA's beneficiaries, could we project it for government costs? ”

- **Problem:** Little is known about the case mix of privately insured patients who are hospitalized in Chile's public network, and whether they should be monitored as a distinct group.
- **Population:** Hospital discharges (egresos) from public hospitals coded with IR-GRD and published by FONASA as annual GRD files.
- **Place:** Chile's public hospital network.
- **Period:** Discharge files 2020, 2021, 2022, 2023, 2024.
- **Proposed use:** Descriptive evidence to support a proposed decision by a public-network management analyst.
- **Policy objective:** Reduce inequities and assure quality of care (ENS 2021 - 2030).

Data quality checks (Five Cs)

1

Consistent

Encoding differs by year (latin1 vs. UTF-16 for 2022/2023) — verified via header inspection, not assumed.

2

Clean

1 malformed line found out of 932,840 rows (2022) — identified and safely excluded.

3

Conformed

2022 “EXTERNO” file confirmed to represent the same population as other years, via PREVISION distribution comparison.

4

Current

Data comes from the last 6 years of hospitals.

5

Comprehensive

Data contained a full dive into the database. We had to select what was relevant for our study.

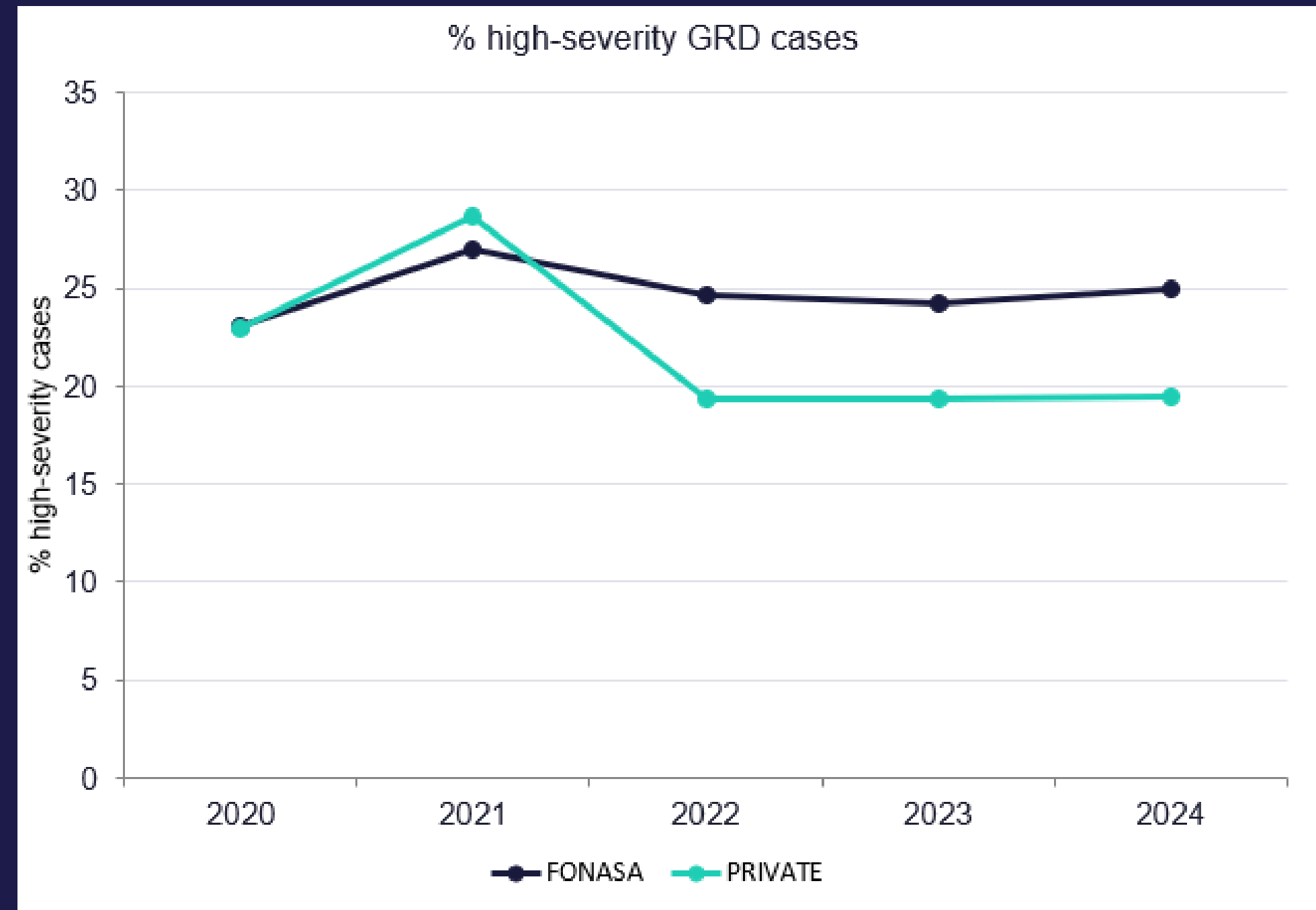
How the KPIs work

KPI 1 — % high-severity cases	KPI 2 — Average GRD weight	KPI 3 — % urgent admission
$\frac{n(\text{SEVERIDAD} = 3)}{n(\text{valid discharges})}$ × 100, by insurance group and year Level 3 is the highest severity the GRD grouper assigns — not a cutoff we chose.	$\frac{\Sigma \text{ PESO}}{n(\text{valid discharges})}$ Simple average of the relative case-mix weight assigned to each discharge.	$\frac{n(\text{TIPO_INGRESO} = \text{URGENCIA})}{n(\text{valid discharges})}$ × 100, by insurance group and year Category name verified against the real data before filtering, not assumed.

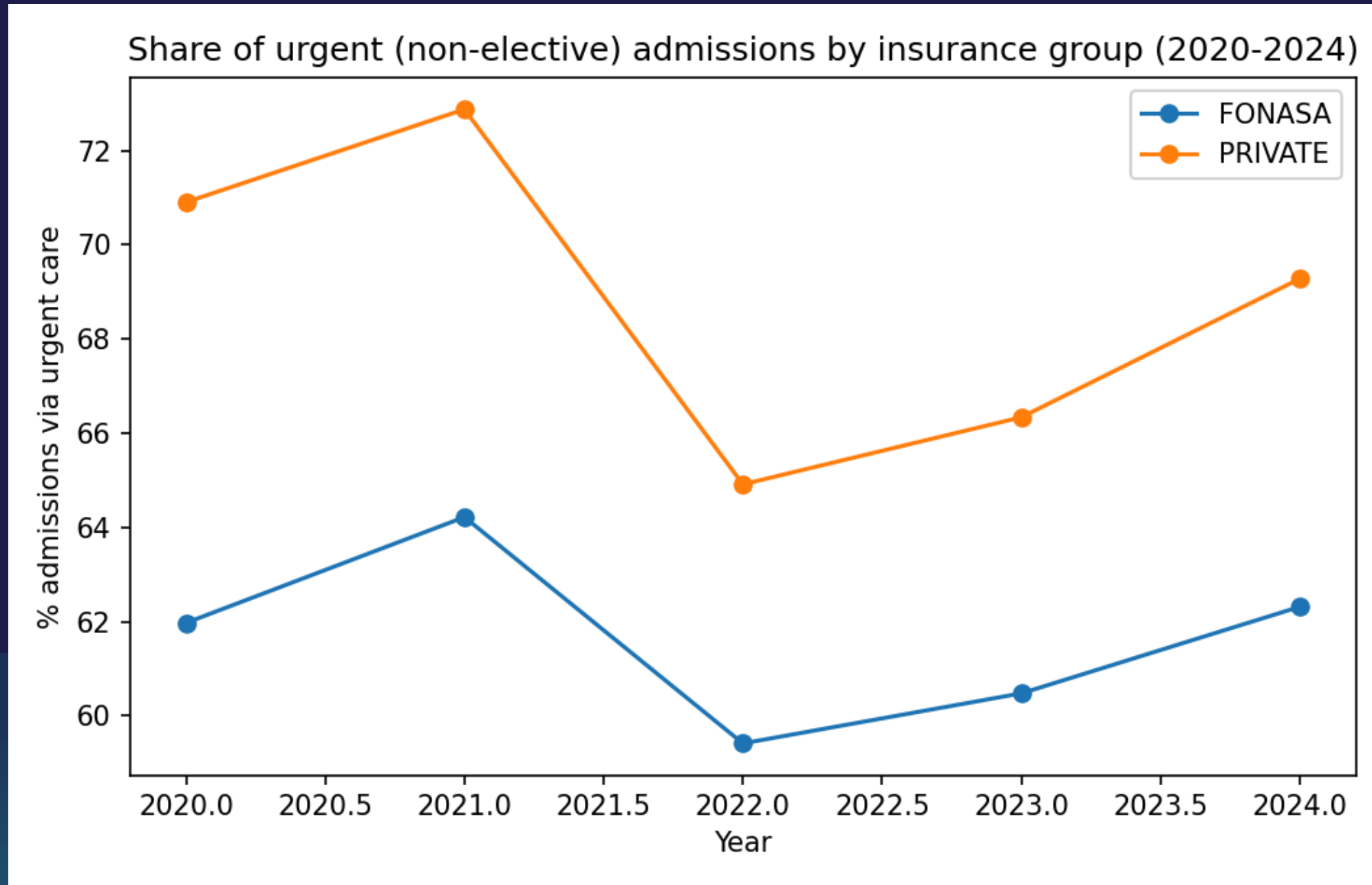
Severity gap by insurance group (2020-2024)

- FONASA and PRIVATE are nearly equal in 2020 (~23%).
- Both spike sharply in 2021 (PRIVATE up to ~28.7%).
- PRIVATE drops and stays ~19-20% from 2022-2024, while FONASA stays ~24-27%.

2021 is treated as an isolated anomaly (fewer postponed elective surgeries that year), not a two-year pandemic pattern — see Slide 6.



Urgent admission share by insurance group (2020-2024)



The most robust finding

- PRIVATE is consistently higher than FONASA in all 5 years, including 2020 and 2021.
- Unlike the severity KPI, this pattern is not affected by the 2021 anomaly.
- Supports the interpretation that privately-insured patients mostly reach the public network through emergencies, not routine or elective care.

Future ML / dashboard potential

ML task

- Predict / classify public hospitals at risk of rising high-severity case load.
- Unit of analysis: hospital × month.
- Inputs: historical PESO / SEVERIDAD distribution, insurance mix, admission-type mix, all available at decision time. Possible opportunity upon importing and locating CENSO data.

To check before C2: sample size per hospital-month; whether Establecimientos-level attributes add predictive value.

Dashboard

- Proposed user: a FONASA public-network management analyst.
- Decision supported: case-load balance and risk-compensation discussions between subsystems.
- Core indicators: case-mix weight and admission-type split by insurance group, tracked over time and by hospital.

Grounded directly in the EDA findings from Slides 5-6.

Conclusions and next steps

Privately-insured patients treated in the public network show lower severity and higher urgent-admission rates than FONASA beneficiaries.

Privately-insured patients appear to treat the public network as a fallback for minor, unplanned incidents — not a substitute for their own care. They almost never enter through scheduled/elective admission, but the cases they do bring in tend to be less clinically severe than FONASA's — suggesting that when something is genuinely serious, it is more likely resolved within their own private network instead of showing up in this data at all.

Cannot Conclude

- A full public-vs-private system comparison, private-hospital GRD data isn't publicly available.
- The thesis above as fact — it is our interpretation, not a tested causal claim; we cannot observe what happens in the private network to confirm it.